EX. NO: 03

DATE: 29/07/2026

IMPLEMENT MINIMAX ALGORITHM

AIM:

To use the MINIMAX algorithm to evaluate the optional move in a game tree.

PYTHON CODE:

```
def minimax(depth, nodeIndex, isMax, scores, h):
    if depth == h:
        return scores[nodeIndex]
    if isMax:
        return max(minimax(depth + 1, nodeIndex * 2, False, scores, h),
                   minimax(depth + 1, nodeIndex * 2 + 1, False, scores, h))
    else:
        return min(minimax(depth + 1, nodeIndex * 2, True, scores, h),
                   minimax(depth + 1, nodeIndex * 2 + 1, True, scores, h))
scores = [3, 5, 6, 9, 1, 2, 0, -1]
h = 3
print("Optimal value:", minimax(0, 0, True, scores, h))
```

OUTPUT:

```
Optimal value: 5
```

RESULT:

Thus the program of using the MINIMAX algorithm to evaluate the optimal move in a game tree was executed successfully.